Example 08 from Siljac book

Consider the system having the dynamical model

$$\dot{x} = Ax + B_1 u_1 + B_2 u_2 + B_3 u_3 + B_4 u_4$$

$$y_i = C_i x, i = 1,2,3,4$$

where the system matrices are specified in the file.

Problem:

1. Generate the system matrices (both continuous-time and discrete-time, the latter with a sampling time $h = 0.1$ s). Perform the following analysis:
    a. Compute the eigenvalues and the spectral abscissa of the (continuous-time) system. Is it open-loop asymptotically stable?
    b. Compute the eigenvalues and the spectral radius of the (discrete-time) system. Is it open-loop asymptotically stable?
2. For different control structures (i.e., centralized, decentralized, and different distributed schemes) perform the following actions
    a. Compute the continuous-time fixed modes
    b. Compute the discrete-time fixed modes
    c. Compute, if possible, the CONTINUOUS-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
    d. Compute, if possible, the DISCRETE-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
    e. Analyze the properties of the so-obtained closed-loop systems (e.g., stability, eigenvalues) and compute the closed-loop system trajectory (generated both in continuous-time and in discrete-time) of variable x_1, starting from a common random initial condition.